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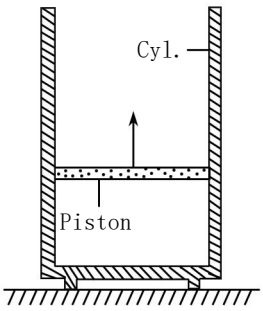
27th National High School Physics Competition: Final Problems

I. (25 points) Fill in the blanks.

1. A uniform thin rubber ring has mass M, stiffness k and natural radius R. Thrown aloft, it stays circular in steady motion. Ignore gravity. It translates and rotates uniformly at ω about its central normal axis. Its radius is _____.

2. A pigeon's whistle has frequency f. Its top flight speed is u. The Doppler effect makes observed frequencies range from _____ to _____. (Sound travels at V.)

3. A vertical insulated cylinder has mass M and inner area A, with a fixed amount of ideal gas at T_0, sealed by a piston. The piston is insulated too. Neglect its mass and friction. Atmospheric pressure is p_0; gravity is g. Pull the piston up slowly. At the opening, the cylinder just leaves the ground. In a slow adiabatic change pV^γ is fixed; p is pressure, V is volume. γ is constant. At the opening, the gas temperature is _____.



4. Use two significant digits. Electron microscopes replace light with electrons to image objects. To resolve $1.0\times10^{-10}\,\mathrm{m}$ (atomic size), the wavelength cannot be larger. Thus the minimum electron speed is _____. To resolve $1.0\times10^{-12}\,\mathrm{m}$ structures, the minimum speed is _____. The required accelerating voltage is _____. The electron has rest mass $m_e=9.1\times10^{-31}\,\mathrm{kg}$ and charge $e=-1.6\times10^{-19}\,\mathrm{C}$. Planck's constant is $h=6.7\times10^{-34}\,\mathrm{J\cdot s}$; the speed of light is $c=3.0\times10^{8}\,\mathrm{m\cdot s^{-1}}$.

II. (20 points) As shown, a point-mass block slides from a platform down a slope onto a long horizontal belt at speed v that carries it far away. The incline's height is $h=2.0\,\mathrm{m}$, horizontal run $L=4.0\,\mathrm{m}$, belt width $d=2.0\,\mathrm{m}$, speed $v=3.0\,\mathrm{m/s}$. Friction is $\mu_1=0.30$. The block starts from rest at the top; while sliding, its velocity is normal to belt motion. A tiny circular arc joins the incline and belt, preserving its speed at the junction. The gravitational acceleration is $g=10\,\mathrm{m/s^2}$.

1. What is the minimum block-belt friction coefficient μ_2 to keep the block from sliding off the belt?

2. A DC motor with a speed regulator drives the belt. The connected supply has emf